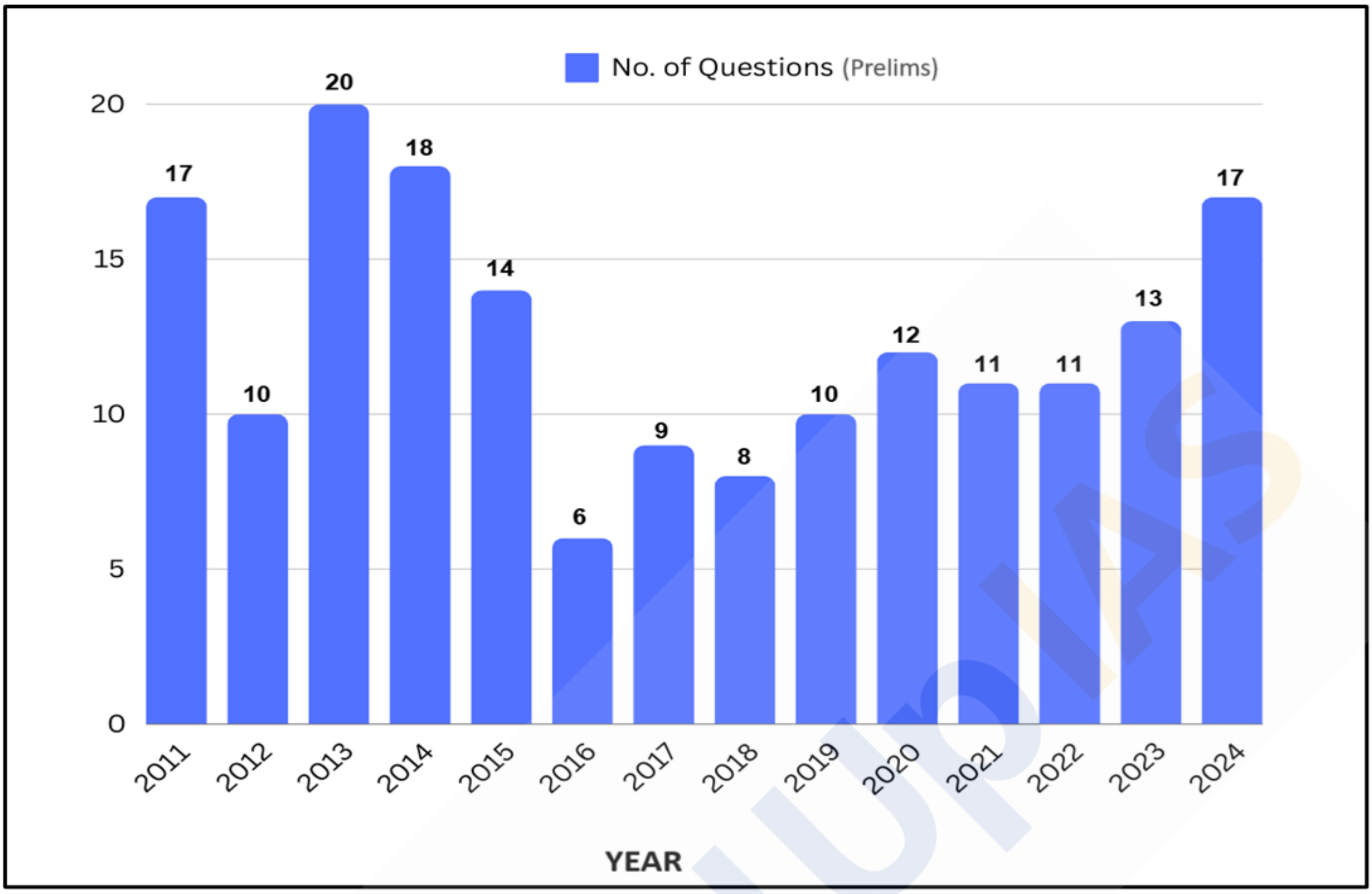
GS FOUNDATION 2024-25

GEOGRAPHY - 10

Water in the Atmosphere-IV

TABLE OF CONTENTS

Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10

Thunderstorm:

Thunderstorms are intense **local storms characterized by rapid upward movement of air** (updraft) **and heavy precipitation, including rainfall, hailstorm, and squall with thunder and lightning.** They are considered tertiary atmospheric circulation and differ from cyclones, which are circular storms with winds blowing towards the center. They occur in **both tropical and temperate** regions.

Characteristics of Thunderstorms:

- **Powerful Local Storms -** Characterized by swift **updrafts of air** from a central point.
- **Thermodynamic Machines - Potential energy is transformed into kinetic energy** through the latent heat of condensation and fusion, providing momentum to the storm.
- **Cellular Structure -** Consist of several convective cells with vertical movement of moist air.
- **Mesoscale Convective Complex -** Sometimes, **multiple thunderstorms** coalesce to form powerful, **large storms.**
- **Convective Cells -** Single-cell thunderstorms are less violent, while multi-cell storms become enormous and disastrous.
- **Tropical and Middle Latitude Thunderstorms -** Tropical thunderstorms involve **heavy downpour, thunder, and lightning.** Middle latitude storms include hail and squalls besides heavy precipitation, thunder, and lightning.
- **Daily Occurrence in Tropics - Frequent over land surfaces** due to intense convection, **less frequent over oceans.**
- **Short-Duration -** Generally last **for an hour or two,** though some can persist for several hours.
- **Global Frequency -** Annually, around 16 million thunderstorms occur globally, with an **average daily occurrence of 2000 storms.**
- **Downburst -** Strong thunderstorms are marked by a powerful downward movement of air, called a **downburst.**

Structure of Thunderstorms:

A fully developed thunderstorm consists of **multiple convective cells, usually ranging from five to eight.** Each cell features a strong updraft of air and may last from one hour to ten hours. The updraft is like a chimney, allowing **cooler air to descend and compensate for the rising warm air.** Thunderstorms progress through three stages of development and dissipation:

Cumulus Stage:

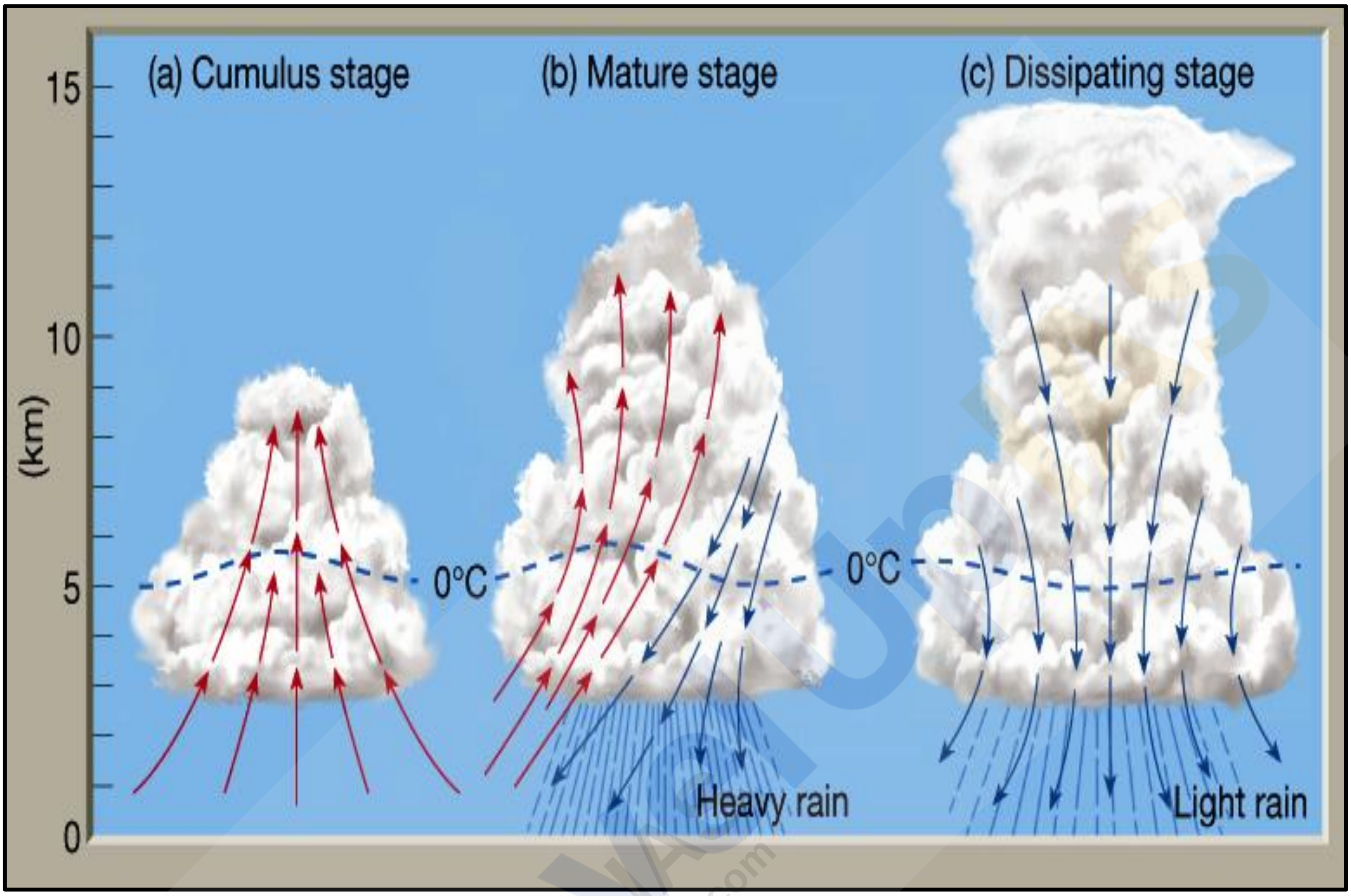
- This initial stage, also known as the **youth stage,** occurs when the **ground surface heats up significantly,** causing the **moist air in contact with it to rise rapidly.** The air **expands and cools** at the dry adiabatic lapse rate of 10°C per 1,000 meters until it reaches its condensation level, forming clouds. As the updraft strengthens, **cumulus clouds develop into thicker cumulonimbus clouds.**

Mature Stage:

- The second stage features both **upward** (updraft) and **downward (**downdraft) **air movements.** During this stage, intense precipitation occurs from the **cumulonimbus clouds,** accompanied by **thunder and lightning.** The storm reaches its peak intensity with **maximum cloud-to-ground lightning activity** and **torrential rainfall.**

Dissipating Stage:

- This final stage is marked by the **cessation of strong updrafts** and the **dominance of downdrafts.** The storm's clouds spread out, forming an umbrella-like shape and eventually transforming into **altostratus and cirrostratus clouds,** which are **rainless.** The loss of vertical air movement and the stabilization of the atmosphere lead to the **storm's dissipation.**



Conditions for Thunderstorm Development:

Intrutive thinking
→ bhai kuch Rain related Jada
→ Toh Moisture Jada ho in air
→ Lapse rate > than adibatic rate

- **Atmospheric Instability -** Conditional or convective instability is necessary.
- **Lapse Rate -** The normal lapse rate must be greater than the adiabatic rate.
- **Warm, Moist, and Rising Air -** Essential for initiating storms.
- **Moisture Supply -** Abundant moisture from the ground to the air is crucial.
- **Cloud Thickness -** Sufficient thickness between the condensation level (cloud base) and icing level.
- **Latent Heat Supply -** Enough latent heat of condensation and fusion to convert potential energy into kinetic energy.
  → Rising air
- **Trigger Effects -**
  - **Insolation Heating -** Ground surface heating causing **convective currents.**
  - **Orographic Lifting -** Forced ascent of moist air **by mountain barriers**.
  - **Cyclonic Fronts -** Forced ascent along cyclonic fronts (applicable in middle latitudes).

Types of Thunderstorms:

Based on Origin:

Convectional Thunderstorms:

- Convectional thunderstorms form due to **intense heating of the ground during summer,** causing warm air to rise rapidly and create thunderstorms.

Orographic Thunderstorms:

- These thunderstorms occur when warm, moist air is **forcefully uplifted over a mountain barrier,** forming cumulonimbus clouds and resulting in heavy precipitation on the windward side. Orographic cloud bursts are common in regions like **Jammu and Kashmir, Cherrapunji, and Mawsynram.**

Frontal Thunderstorms:

- Frontal thunderstorms occur along cold fronts, where the **meeting of cold and warm air masses** leads to storm formation and associated weather changes.

Based on Life Span:

Single-Cell Thunderstorms:

- Single-cell thunderstorms are **small, brief, and weak storms that grow and die within an hour,** typically driven by heating on a summer afternoon. They produce brief heavy rain and lightning, commonly referred to as **'Mango Showers'** in Kerala and **'Blossom showers'** in Karnataka.

Multi-Cell Thunderstorms:

- A multi-cell storm consists of **multiple updrafts forming along the leading edge of rain-cooled air.** Individual cells **last 30 to 60 minutes,** but the system can last many hours, producing hail, strong winds, **brief tornadoes,** and flooding.

Supercell Thunderstorms:

- Supercell thunderstorms are **highly organized, long-lived storms with a rotating updraft.** They are the source of most large and violent tornadoes.

Lightning and Thunder:

Lightning occurs due to the movement of water vapor upward in cumulonimbus clouds, causing condensation and release of latent heat. As water droplets freeze into ice crystals and fall, **collisions release electrons, creating ionization.** This results in a **charge separation within the cloud, leading to an electrical potential difference of 10^9 to 10^{10} volts.** The subsequent **current flow produces heat, causing the air column to expand and generate shock waves,** resulting in **'thunder.'** Lightning can also strike the Earth, often targeting tall objects like trees and buildings.

Lightning Deaths:

Lightning strikes in India have **increased over the past 20 years,** especially near the **Himalayas.** Most lightning fatalities occur due to ground currents, where electrical energy spreads laterally from a struck object. Wet ground and **conducting materials like metal can increase the danger.**

Prediction and Precautions:

Predicting the precise location and timing of thunderstorms is challenging. It is advisable to move indoors during a storm and **avoid electrical fittings, wires, metal, and water.** Shelter under trees or lying flat on the ground should be avoided.

The World's Most Electric Place:

Lake Maracaibo in Venezuela experiences the most lightning activity on Earth, especially in October, with **28 lightning flashes every minute.** This phenomenon, known as the **Beacon of Maracaibo,** is caused by the unique topography and climatic conditions of the region.

Types of Lightning Strikes:

- **Direct Strike -** Occurs most often in open areas.
- **Side Flash -** Occurs when lightning strikes a taller object and some current jumps to a nearby victim.
- **Ground Current -** Energy travels outward along the ground surface from a struck object.
- **Conduction -** Lightning can travel long distances in wires or metal surfaces.

Features of Lightning:

Positive charges accumulate at higher and lower altitudes, while **larger cloud particles have a negative polarity.** Most discharges occur within the cloud, with a minority between the cloud and ground. **Lightning creates plasma, resulting in shock waves heard as thunder** once they slow to the speed of sound.

===== XXXXX =====